

Week 2 – Managing Files and Getting Help

Topics covered

- File-system hierarchy
- Absolute and relative paths
- Viewing the current directory
- Listing files and directories
- Creating directories and files
- Copying, moving and deleting files
- Using manual pages
- Searching manual pages

The Linux file system is arranged as an inverted tree. The root directory, written as `/`, is at the top. All other files and directories are located below it.

1. Important Linux directories

Directory	Purpose
<code>/</code>	Top of the Linux file-system hierarchy
<code>/boot</code>	Contains files required during startup
<code>/dev</code>	Contains device files representing hardware
<code>/etc</code>	Contains system configuration files
<code>/home</code>	Contains regular users' home directories
<code>/root</code>	Home directory of the root user

<code>/run</code>	Contains runtime data created since boot
<code>/tmp</code>	Stores temporary files
<code>/usr</code>	Contains installed software and program data
<code>/var</code>	Contains changing data such as logs and website files

Examples:

`ls /`

Lists the main directories under the root directory.

`ls /etc`

Lists configuration files and directories stored under `/etc`.

2. Absolute and relative paths

An **absolute path** gives the complete location of a file or directory. It always begins with `/`.

Example:

`/var/log`

A **relative path** is based on your current working directory.

Example:

`Documents/report.txt`

If your current directory is `/home/student`, the full path would be:

`/home/student/Documents/report.txt`

3. Checking the current directory

Use:

`pwd`

`pwd` means **Print Working Directory**.

Example output:

`/home/student`

4. Changing directories

Move to `/etc`:

`cd /etc`

Return to your home directory:

`cd`

You can also use:

`cd ~`

Move to the parent directory:

`cd ..`

Return to the previous directory:

`cd -`

Move into a subdirectory:

`cd Documents`

5. Special path symbols

Symbol	Meaning
--------	---------

/	Root directory
.	Current directory
..	Parent directory
~	Current user's home directory
-	Previous working directory

Examples:

ls .

Lists the current directory.

ls ..

Lists the parent directory.

ls ~

Lists your home directory.

6. Listing files

Basic listing:

ls

Detailed listing:

ls -l

Show hidden files:

ls -a

Show hidden files in detailed format:

ls -la

List the directory itself instead of its contents:

ls -ld /etc

List multiple locations:

ls /etc /var/log

7. Understanding **ls -l** output

Example:

```
-rw-r--r--. 1 student student 120 Jul 23 10:00 notes.txt
```

Field	Meaning
-	File type
rw-r--r--	File permissions
1	Number of hard links
student	File owner
student	Group owner

120 File size

Jul 23 Last modification time
10:00

notes.txt File name

Common file-type symbols:

Symbol	Meaning
--------	---------

d	Directory
---	-----------

-	Regular file
---	--------------

l	Symbolic link
---	---------------

Examples:

drwxr-xr-x

The first character is d, so it is a directory.

-rw-r--r--

The first character is -, so it is a regular file.

lrwxrwxrwx

The first character is l, so it is a symbolic link.

8. Creating directories

Create one directory:

```
mkdir practice_dir
```

Create several directories:

```
mkdir dir1 dir2 dir3
```

Create a directory together with missing parent directories:

```
mkdir -p project/docs/reports
```

Without `-p`, the command fails if the parent directories do not already exist.

Verify:

```
ls -ld practice_dir
```

9. Creating empty files

Use:

```
touch testfile.txt
```

If the file does not exist, `touch` creates an empty file.

If the file already exists, `touch` updates its modification timestamp.

Verify:

```
ls -l testfile.txt
```

10. Copying files and directories

Copy a file:

```
cp testfile.txt backup.txt
```

This creates `backup.txt` with the same contents as `testfile.txt`.

Copy a file to another directory:

```
cp testfile.txt /tmp
```

Copy a directory and everything inside it:

```
cp -r practice_dir practice_backup
```

The `-r` option means recursive and is required when copying directories.

Interactive copy:

```
cp -i testfile.txt backup.txt
```

The `-i` option asks for confirmation before overwriting an existing file.

11. Moving and renaming files

Move a file:

```
mv backup.txt /tmp
```

Rename a file:

```
mv testfile.txt newname.txt
```

Move and rename at the same time:

```
mv newname.txt /tmp/final.txt
```

Interactive move:

```
mv -i file1.txt file2.txt
```

This asks before overwriting an existing destination file.

12. Deleting files and directories

Delete a file:

```
rm testfile.txt
```

Ask for confirmation:

```
rm -i testfile.txt
```


Delete an empty directory:

```
rmdir practice_dir
```

Delete a directory and its contents:

```
rm -r practice_dir
```

Force recursive deletion:

```
rm -rf practice_dir
```

Use `rm -rf` carefully because deleted files are not normally moved to a recycle bin.

13. Using wildcards

Wildcards allow one command to work with several matching file names.

Wildcard	Meaning
<code>*</code>	Matches zero or more characters
<code>?</code>	Matches exactly one character
<code>[abc]</code>	Matches one listed character
<code>[a-z]</code>	Matches one character from a range
<code>[!abc]</code>	Matches a character not listed

Examples:

List all `.txt` files:

ls *.txt

List files beginning with **report**:

ls report*

List names containing exactly one character after **file**:

ls file?

List files beginning with **a**, **b** or **c**:

ls [abc]*

Be careful when using wildcards with **rm**:

rm *.txt

This deletes every matching **.txt** file in the current directory.

14. Getting command help

Basic command help:

ls --help

This gives a compact summary of available options.

Open the manual page:

man ls

Useful controls inside a manual page:

Key	Purpose
Spacebar	Move forward one page

b Move back one page

/word Search for a word

n Next matching result

Shift+n Previous matching result

q Quit

15. Manual-page sections

Linux manual pages are divided into sections.

Section	Content
1	User commands
2	System calls
3	Library functions
4	Special device files

- 5 File formats and configuration files
- 6 Games
- 7 Miscellaneous
- 8 System-administration commands
- 9 Kernel routines

Open the command manual for `passwd`:

```
man 1 passwd
```

Open the manual describing the `/etc/passwd` file format:

```
man 5 passwd
```

These are different manual pages even though they have the same name.

16. Searching for manual pages

Search by keyword:

```
man -k password
```

This lists manual pages whose names or descriptions contain the word `password`.

The following command provides similar results:

```
apropos password
```

If `man -k` produces no results, the manual-page database may need to be updated by an administrator:

```
sudo mandb
```

